

Cybersecurity Core Terms and Concepts Study Guide

Cybersecurity (also known as security) is the practice of ensuring confidentiality, integrity, and availability of information by protecting networks, devices, people, and data from unauthorized access or criminal exploitation. This guide covers key foundational terms and concepts used by security professionals to identify and manage digital threats.

■ Security Management & Strategy

Compliance

Compliance means following both the **internal rules** of a company and the **external laws** set by governments. It is like driving at the speed limit and wearing a seatbelt; doing so helps the organization avoid expensive fines, legal trouble, and security breaches.

Security Frameworks

A security framework is a **blueprint or master plan** used to build a strong defense. It provides step-by-step guidelines on how to protect data, respect privacy, and minimize digital risks.

Security Controls

Security controls are the specific **safeguards or tools** put in place to block specific risks. If a security framework says 'you must protect your doors,' the security control is the actual physical lock or the password screen you use to keep hackers out.

Security Posture

Security posture is an organization's overall **readiness to defend itself**. It measures how well a company can protect its data and how quickly it can react to a new cyberattack. A 'strong posture' means the company is highly prepared and faces a much lower risk of being hacked.

■ The People Involved (Analysts & Attackers)

Cybersecurity Analyst

A security analyst is a digital **security guard** for a company's computer networks. Their main jobs are to:

- Watch networks 24/7 to spot unauthorized break-ins.
- Build defense strategies to keep the organization safe.
- Research new hacking trends to stay one step ahead of criminals.
- Stop cyber incidents before they even happen.

Threat Actor

A threat actor (also called a malicious attacker) is **any person or group** that tries to cause a security risk. They are the criminals trying to break into computers, applications, networks, or databases.

Internal Threat

An internal threat is a security danger that comes from **inside the organization**. This could be a current worker, a former employee, or a trusted business partner. Internal threats happen in two ways:

- **Accidental Threat:** A regular employee who makes an honest mistake, like accidentally clicking a bad link in a phishing email.
- **Intentional Threat:** A rogue worker who deliberately steals secret company data or accesses systems they are not allowed to see.

■ ■ Areas of Protection

Network Security

Network security is the practice of locking down a company's **entire digital infrastructure**. It focuses on protecting all the connected data, services, software systems, and physical devices (like routers and computers) living inside the organization's private network.

Cloud Security

The **cloud** is a massive network of remote computers stored in giant physical buildings called data centers, which you access over the internet. **Cloud security** is the specific job of making sure these remote systems are set up correctly so that hackers cannot access them, limiting entry strictly to allowed users.

■ Automated Tools

Programming

Programming is the process of writing **step-by-step instructions** that a computer can follow. Security analysts use programming to make their jobs easier by handling three major tasks:

- **Automation:** Making computers automatically handle boring, repetitive jobs (like checking a list of millions of websites to see which ones are dangerous).
- **Reviewing Web Traffic:** Scanning massive amounts of internet data flowing into the company to see if anything looks unusual.
- **Alerting:** Setting up the system to automatically trigger a loud alarm or send an email the exact second suspicious activity is detected.